

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Trigonometrical Ratios

Q1

If $y = \sqrt{\frac{2(\tan \alpha + \cot \alpha)}{1 + \tan^2 \alpha} + \frac{1}{\sin^2 \alpha}}$ when $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$ then find $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$

यदि $y = \sqrt{\frac{2(\tan \alpha + \cot \alpha)}{1 + \tan^2 \alpha} + \frac{1}{\sin^2 \alpha}}$ जब $\alpha \in \left(\frac{3\pi}{4}, \pi\right)$ तब $\alpha = \frac{5\pi}{6}$ पर $\frac{dy}{d\alpha}$ का मान होगा—

(1) 4

(2) 2

(3) 3

(4) -4

7th Jan Morning

Sol

(1)

$$y = \sqrt{\frac{2\cos^2 \alpha}{\sin \alpha \cos \alpha} + \frac{1}{\sin^2 \alpha}} = \sqrt{2\cot \alpha + \operatorname{cosec}^2 \alpha} = |1 + \cot \alpha| = -1 - \cot \alpha$$

$$\frac{dy}{d\alpha} = \operatorname{cosec}^2 \alpha \Rightarrow \left(\frac{dy}{d\alpha}\right)_{\alpha = \frac{5\pi}{6}} \text{ will be } = 4$$

02

Let $\frac{\sqrt{2} \sin \alpha}{\sqrt{1 + \cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1 - \cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$ where $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$. Then $\tan(\alpha + 2\beta)$ is equal to

8th Jan Evening

Sol

(1)

$$\frac{\sqrt{2} \sin \alpha}{\sqrt{2} \cos \alpha} = \frac{1}{7} \quad \text{and तथा} \quad \frac{\sqrt{2} \sin \beta}{\sqrt{2}} = \frac{1}{\sqrt{10}}$$

$$\tan \alpha = \frac{1}{7}$$

$$\sin \beta = \frac{1}{\sqrt{10}}$$

03

Value of $\cos^3 \frac{\pi}{8} \cos \frac{3\pi}{8} + \sin^3 \frac{\pi}{8} \sin \frac{3\pi}{8}$ is

$\cos^3 \frac{\pi}{8} \cos \frac{3\pi}{8} + \sin^3 \frac{\pi}{8} \sin \frac{3\pi}{8}$ का मान है—

(1) $\frac{1}{2\sqrt{2}}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

9th Jan Morning

Sol

(1)

$$\cos^3 \frac{\pi}{8} \left[4 \cos^3 \frac{\pi}{8} - 3 \cos \frac{\pi}{8} \right] + \sin^3 \frac{\pi}{8} \left[3 \sin \frac{\pi}{8} - 4 \sin^3 \frac{\pi}{8} \right]$$

$$= 4 \cos^6 \frac{\pi}{8} - 4 \sin^6 \frac{\pi}{8} - 3 \cos^4 \frac{\pi}{8} + 3 \sin^4 \frac{\pi}{8}$$

$$= 4 \left[\left(\cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8} \right) \right] \left[\left(\sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) \right] - 3 \left[\left(\cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8} \right) \right]$$

$$= \cos \frac{\pi}{4} \left[4 \left(1 - \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) - 3 \right] = \frac{1}{\sqrt{2}} \left[1 - \frac{1}{2} \right] = \frac{1}{2\sqrt{2}}$$

04

Let $x = \sum_{n=0}^{\infty} (-1)^n (\tan \theta)^{2n}$ and $y = \sum_{n=0}^{\infty} (\cos \theta)^{2n}$ where $\theta \in (0, \pi/4)$, then

माना $x = \sum_{n=0}^{\infty} (-1)^n (\tan \theta)^{2n}$ तथा $y = \sum_{n=0}^{\infty} (\cos \theta)^{2n}$ जहाँ $\theta \in (0, \pi/4)$, तब

(1) $x(y+1) = 1$

(2) $y(1-x) = 1$

(3) $y(x-1) = 1$

(4) $y(1+x) = 1$

9th Jan Evening

Sol

(2)

$$y = 1 + \cos^2\theta + \cos^4\theta + \dots\dots$$

$$\Rightarrow y = \frac{1}{1 - \cos^2\theta} \Rightarrow \frac{1}{y} = \sin^2\theta$$

$$x = \frac{1}{1 - (-\tan^2\theta)} = \frac{1}{\sec^2\theta}$$

$$\Rightarrow \cos^2\theta = x$$

$$\Rightarrow \frac{1}{y} + x = 1$$

$$\Rightarrow y(1 - x) = 1$$



mathongo

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